**Pre Analysis Plan for #ManyDaughters Project**

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## General Comment

We focus on the first-born child and whether this child is a daughter or not (https://doi.org/10.1111/j.1467-937X.2008.00514.x).
We will do a cross-sectional analysis using the last panel wave available.

## Hypotheses

**H1.** *Having daughters leads to stronger support for gender equality (measured by attitudes).*
Let **X** represent a measure of support for gender equality. Then:
**X (respondents with daughters) > X (respondents without daughters)**

## Variables

- X: pl/plj0620: Demokratie: Frauen haben gleiche Rechte wie Maenner (0=Gehoert auf keinen Fall zur Demokratie, (Skala 0-10))
- Sex of first-born child: pld0175 [scale level 2=daughter]: first child daughter [0/1]
- Controls: gender (sex_h), age (alteri), household income (hl/hlc0005_h)

## Inclusion/exclusion criteria

- No exclusion criteria. All households with observations in the last available panel wave.

## Exact analysis (person level)

*X = beta_0 + beta_1*1[first child daughter] + beta_2*gender + beta_3*age + beta_4*income + error*

## Interpretation of results

A significant positive coefficient of *beta_1* indicates support for the hypothesis.

**H2.** *Having daughters leads to more left-leaning political preferences.*
Let **Y** represent a measure of left-leaning political preferences. Then:
**Y (respondents with daughters) > Y (respondents without daughters)**

**Variables**

- Y: pl/plh0004: Politische Einstellung (0=Ganz links, (Skala 0-10))
- Sex of first-born child: pld0175 [scale level 2=daughter]: first child daughter [0/1]
- Controls: gender (sex_h), age (alteri), household income (hl/hlc0005_h)

**Inclusion/exclusion criteria**

- No exclusion criteria. All households with observations in the last available panel wave.

**Exact analysis (person level)**

*Y = beta_0 + beta_1*1[first child daughter] + beta_2*gender + beta_3*age + beta_4*income + error*

**Interpretation of results**

A significant negative coefficient of *beta_1* indicates support for the hypothesis.

**H3.** *Having daughters leads to more gender equality in the household (measured by behaviors or outcomes).*
Let **Z** represent a measure of gender equality in the household. Then:
**Z (respondents with daughters) > Z (respondents without daughters)**

**Variables**

- Z: Anteil Hausarbeit Mann: Summe Hausarbeit Mann / Summe Hausarbeit Haushalt
    - For every household id: hid: calculate sum Hausarbeit [hausarbeit: pl/pli0012_h: Hausarbeit Std., Samstag, pl/pli0016_h: Hausarbeit Std., Sonntag, pl/pli0043_h: Hausarbeit Std., Werktg.] for each household member by gender (sex_h)
- Sex of first-born child: pld0175 [scale level 2=daughter]: first child daughter [0/1]
- Controls: household income (hl/hlc0005_h)

**Inclusion/exclusion criteria**

- No exclusion criteria. All households with observations in the last available panel wave.

**Exact analysis (household level)**
*Z = beta_0 + beta_1*1[first child daughter] + beta_2*income + error*

**Interpretation of results**
A significant positive coefficient of *beta_1* indicates support for the hypothesis.

**H4.** *Having daughters leads to a larger increase in support for gender equality among men than among women (measured by attitudes).*
Let **X** again represent a measure of support for gender equality. Then:
**[X (men with daughters) − X (men without daughters)] > [X (women with daughters) − X (women without daughters)]**

**Variables**

- X: pl/plj0620: Demokratie: Frauen haben gleiche Rechte wie Maenner (0=Gehoert auf keinen Fall zur Demokratie, (Skala 0-10))
- Sex of first-born child: pld0175 [scale level 2=daughter]: first child daughter [0/1]
- Is man: 1(sex_h=1)
- Controls: gender (sex_h), age (alteri), household income (hl/hlc0005_h)

**Inclusion/exclusion criteria**

- No exclusion criteria. All households with observations in the last available panel wave.

**Exact analysis (person level)**

*X = beta_0 + beta_1*1[is man]*1[first child daughter] + beta_2*1[first child daughter] + beta_3*1[is man] + beta_4*age + beta_5*income + error*

**Interpretation of results**

A significant positive coefficient of *beta_1* indicates support for the hypothesis.